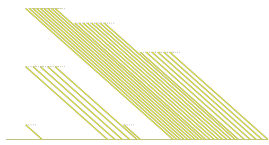


Chapitre 4 : La régression

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1 Avant de commencer

Ce chapitre reprends les codes du chapitre 4 sur **la régression** du livre **Langage R et Statistiques** : Initiation à l'analyse de données publié par les éditions ENI (2^{ème} édition).

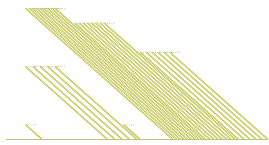
```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.1      ✓ readr      2.2.0
✓ forcats    1.0.1      ✓ stringr    1.6.0
✓ ggplot2    4.0.3      ✓ tibble     3.3.1
✓ lubridate  1.9.5      ✓ tidyr      1.3.2
✓ purrr      1.2.2
— Conflicts — tidyverse_conflicts() —
* dplyr::filter() masks stats::filter()
* dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
  conflicts to become errors
```

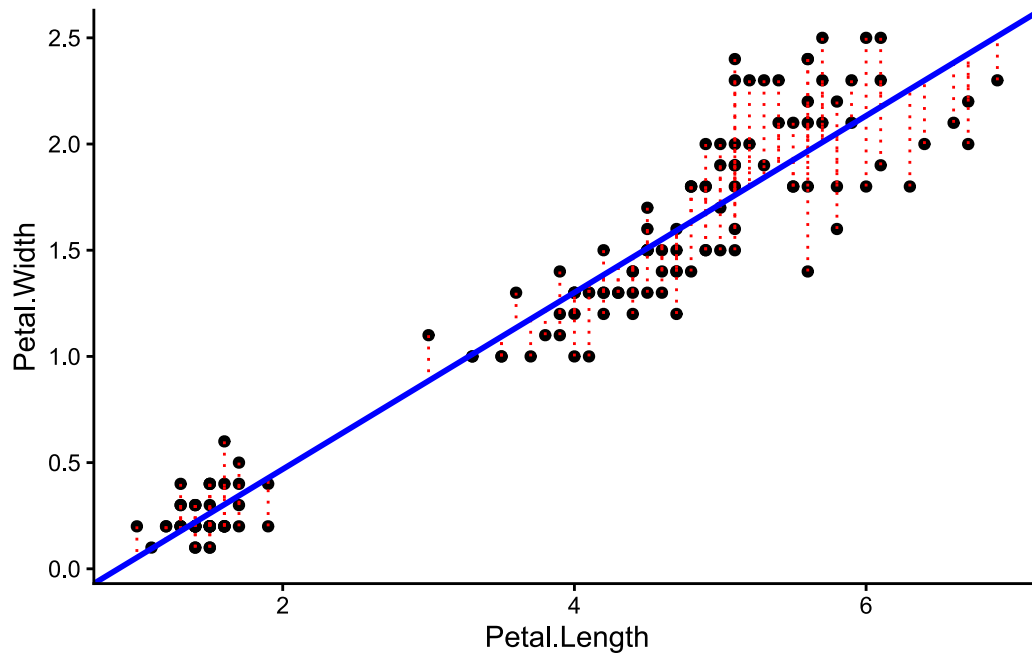
2 Régression linéaire simple

2.1 graphique pour visualiser les moindres carrés

```
ggplot(iris) +
  aes(
    x = Petal.Length,
    y = Petal.Width
  ) +
  geom_point() +
  geom_segment(
    aes(
      x = Petal.Length,
      xend = Petal.Length,
      y = Petal.Width,
      yend = 0.416 * Petal.Length - 0.363
    ),
    linetype = "dotted",
    color = "red"
  ) +
  geom_abline(slope = 0.416, intercept = -0.363, color = "blue", size = 1) +
  theme_classic()
```



Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.



2.2 modèle linéaire regression simple

```
regression_lineaire <- lm(  
  formula = Petal.Width ~ Petal.Length,  
  data = iris  
)
```

```
summary(regression_lineaire)
```

Call:

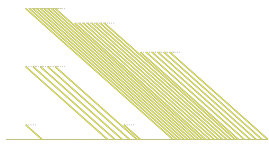
```
lm(formula = Petal.Width ~ Petal.Length, data = iris)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.56515	-0.12358	-0.01898	0.13288	0.64272

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.363076	0.039762	-9.131	4.7e-16 ***



```
Petal.Length 0.415755 0.009582 43.387 < 2e-16 ***
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2065 on 148 degrees of freedom
Multiple R-squared: 0.9271, Adjusted R-squared: 0.9266
F-statistic: 1882 on 1 and 148 DF, p-value: < 2.2e-16
```

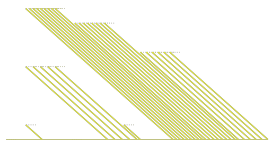
2.3 représentation graphique pour données autre que linéaires

```
purrr::map2(
  .x = c("x1", "x2", "x3", "x4"),
  .y = c("y1", "y2", "y3", "y4"),
  .f = ~ ggplot(anscombe) +
    aes(x = get(.x), y = get(.y)) +
    geom_point() +
    geom_abline(slope = 0.5, intercept = 3, color = "blue") +
    annotate(
      "text",
      x = 14,
      y = 3,
      label = "y ~ 0,5 * x + 3 \n R² = 0.63 \n p-valeur < 0.01"
    ) +
    labs(x = .x, y = .y) +
    coord_cartesian(xlim = c(5, 19), ylim = c(0, 13)) +
    theme_classic()
)|>
purrr::map2(
  .y = c("img/04Rl03N.png", "img/04Rl03bisN.png", "img/04Rl03terN.png",
"img/04Rl03quaN.png"),
  .f = ~ ggsave(plot = .x, filename = .y, width = 6, height = 6, units = "cm")
)
```

```
[[1]]
[1] "img/04Rl03N.png"

[[2]]
[1] "img/04Rl03bisN.png"

[[3]]
[1] "img/04Rl03terN.png"
```



```
[[4]]  
[1] "img/04Rl03quaN.png"
```

```
purrr::map2(  
  .x = c("x1", "x2", "x3", "x4"),  
  .y = c("y1", "y2", "y3", "y4"),  
  .f = ~ lm(get(.y) ~ get(.x), data = anscombe) |> summary()  
)
```

```
[[1]]
```

Call:

```
lm(formula = get(.y) ~ get(.x), data = anscombe)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.92127	-0.45577	-0.04136	0.70941	1.83882

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.0001	1.1247	2.667	0.02573 *
get(.x)	0.5001	0.1179	4.241	0.00217 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.237 on 9 degrees of freedom

Multiple R-squared: 0.6665, Adjusted R-squared: 0.6295

F-statistic: 17.99 on 1 and 9 DF, p-value: 0.00217

```
[[2]]
```

Call:

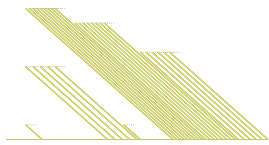
```
lm(formula = get(.y) ~ get(.x), data = anscombe)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.9009	-0.7609	0.1291	0.9491	1.2691

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.001	1.125	2.667	0.02576 *



```
get(.x)          0.500      0.118   4.239  0.00218 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.237 on 9 degrees of freedom
Multiple R-squared:  0.6662,    Adjusted R-squared:  0.6292
F-statistic: 17.97 on 1 and 9 DF,  p-value: 0.002179

[[3]]

Call:
lm(formula = get(.y) ~ get(.x), data = anscombe)

Residuals:
    Min       1Q   Median       3Q      Max
-1.1586 -0.6146 -0.2303  0.1540  3.2411

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.0025     1.1245   2.670  0.02562 *
get(.x)        0.4997     0.1179   4.239  0.00218 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

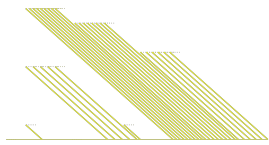
Residual standard error: 1.236 on 9 degrees of freedom
Multiple R-squared:  0.6663,    Adjusted R-squared:  0.6292
F-statistic: 17.97 on 1 and 9 DF,  p-value: 0.002176

[[4]]

Call:
lm(formula = get(.y) ~ get(.x), data = anscombe)

Residuals:
    Min       1Q   Median       3Q      Max
-1.751 -0.831  0.000  0.809  1.839

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.0017     1.1239   2.671  0.02559 *
get(.x)        0.4999     0.1178   4.243  0.00216 **
```



```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.236 on 9 degrees of freedom
Multiple R-squared:  0.6667,    Adjusted R-squared:  0.6297
F-statistic:    18 on 1 and 9 DF,  p-value: 0.002165
```

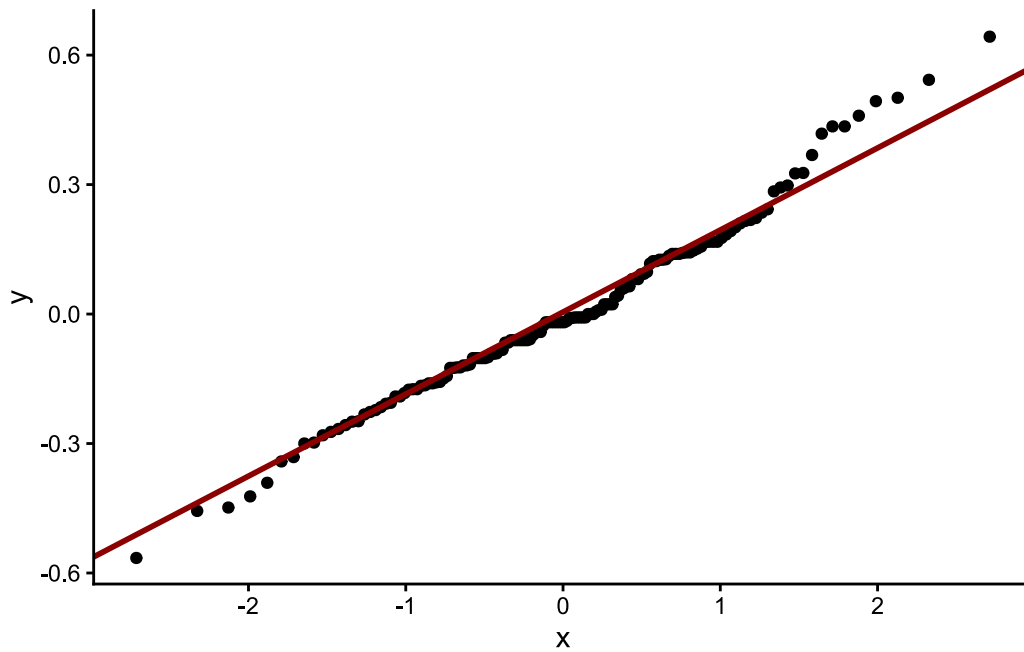
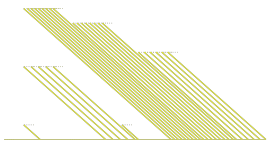
2.4 Normalité des résidus

```
shapiro.test(regression_lineaire$residuals)
```

Shapiro-Wilk normality test

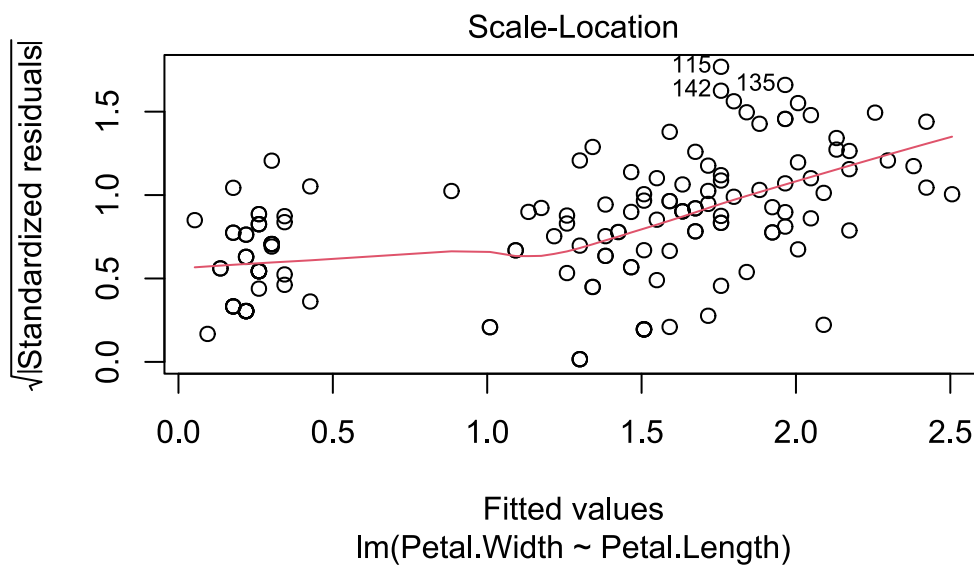
```
data:  regression_lineaire$residuals
W = 0.98378, p-value = 0.07504
```

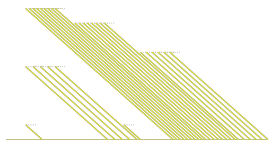
```
tibble(
  residus = regression_lineaire$residuals
) |>
  ggplot() +
  aes(sample = residus) +
  geom_qq() +
  geom_qq_line(color = "dark red", size = 1) +
  theme_classic()
```



2.5 homogéinité

```
plot(regression_lineaire, which = 3)
```

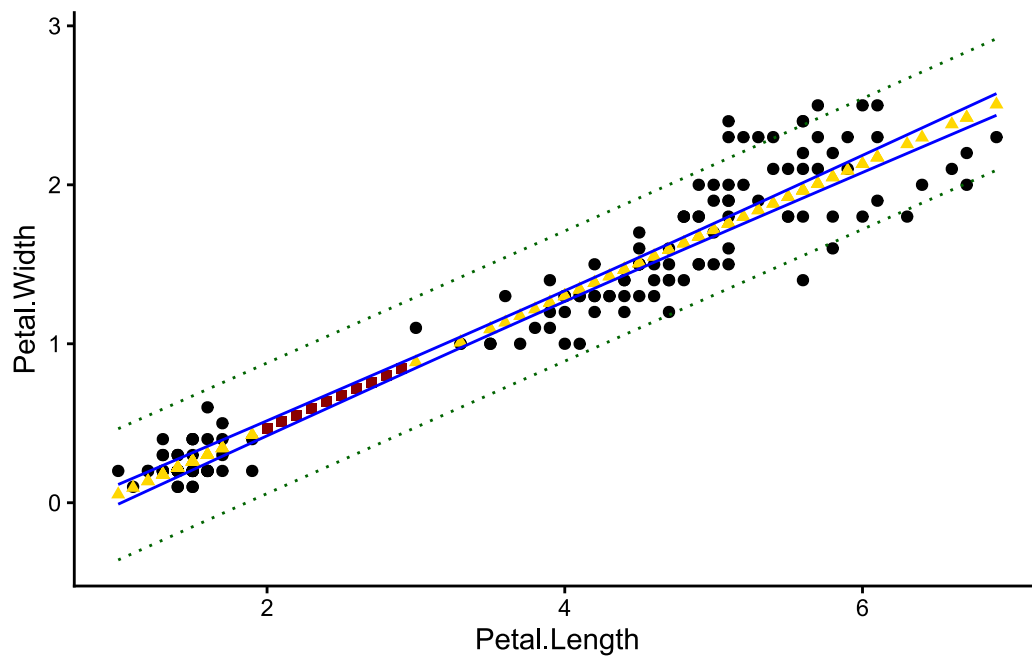
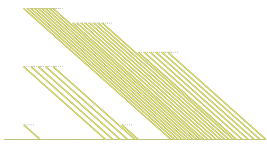




2.6 intervalle de confiance

```
iris|>
  ggplot() +
  aes(
    x = Petal.Length,
    y = Petal.Width
  ) +
  geom_point() +
  geom_point(aes(y = predict(regression_lineaire)), shape = 17, color = "gold") +
  geom_point(
    data = tibble(
      Petal.Length = seq(2, 2.9, 0.1),
      Petal.Width = predict(regression_lineaire, newdata = data.frame(Petal.Length
= seq(2, 2.9, 0.1)))
    ),
    color = "dark red",
    shape = 15
  ) +
  geom_line(aes(y = predict(regression_lineaire, interval = c("confidence"))[,
2]), color = "blue") +
  geom_line(aes(y = predict(regression_lineaire, interval = c("confidence"))[,
3]), color = "blue") +
  geom_line(aes(y = predict(regression_lineaire, interval = c("prediction"))[,
2]), color = "darkgreen", linetype = "dotted") +
  geom_line(aes(y = predict(regression_lineaire, interval = c("prediction"))[,
3]), color = "darkgreen", linetype = "dotted") +
  theme_classic()
```

Warning in predict.lm(regression_lineaire, interval = c("prediction")): les
prédictions sur les données actuelles se réfèrent à des réponses _futures_
Warning in predict.lm(regression_lineaire, interval = c("prediction")): les
prédictions sur les données actuelles se réfèrent à des réponses _futures_



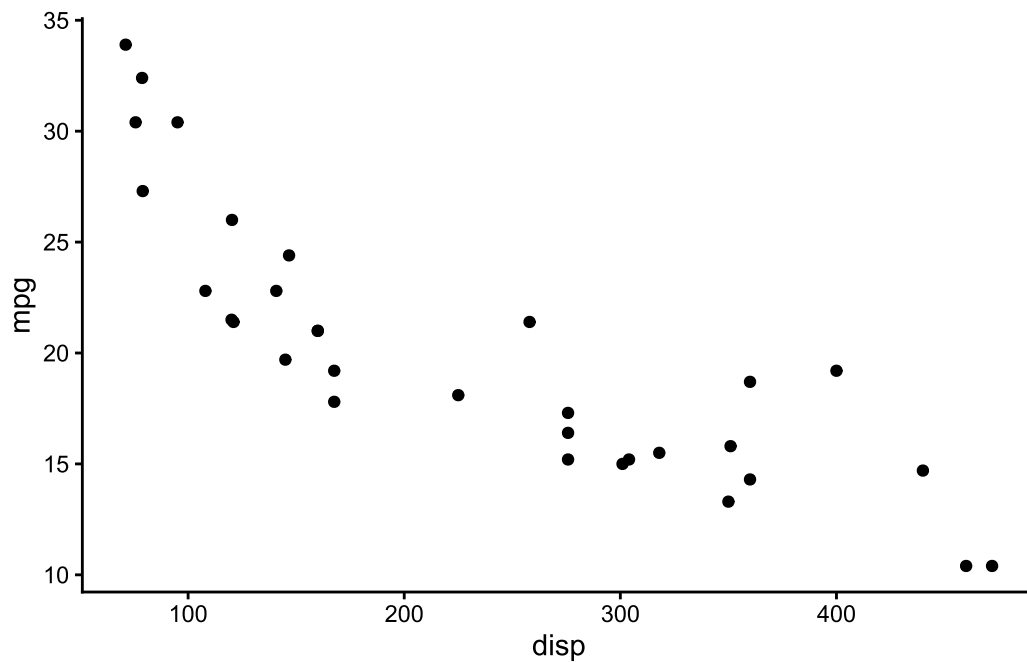
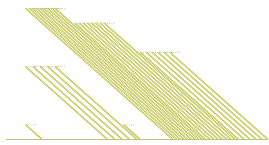
```
ggsave("img/04Rl07N.png", width = 14, height = 7, units = "cm")
```

Warning in predict.lm(regression_lineaire, interval = c("prediction")): les
prédictions sur les données actuelles se réfèrent à des réponses _futures_
Warning in predict.lm(regression_lineaire, interval = c("prediction")): les
prédictions sur les données actuelles se réfèrent à des réponses _futures_

3 Régression polynomiale

3.1 graphique

```
ggplot(mtcars) +  
  aes(y = mpg, x = disp) +  
  geom_point() +  
  theme_classic()
```



3.2 régression linéaire

```
regression_lineaire_cars <- lm(mpg ~ disp, data = mtcars)
summary(regression_lineaire_cars)
```

Call:

```
lm(formula = mpg ~ disp, data = mtcars)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.8922	-2.2022	-0.9631	1.6272	7.2305

Coefficients:

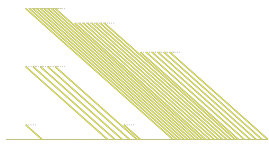
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	29.599855	1.229720	24.070	< 2e-16 ***
disp	-0.041215	0.004712	-8.747	9.38e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.251 on 30 degrees of freedom

Multiple R-squared: 0.7183, Adjusted R-squared: 0.709

F-statistic: 76.51 on 1 and 30 DF, p-value: 9.38e-10



vérification de normalité des résidus

```
shapiro.test(regression_lineaire_cars$residuals)
```

Shapiro-Wilk normality test

```
data:  regression_lineaire_cars$residuals  
W = 0.9271, p-value = 0.03255
```

3.3 régression polynomiale

```
regression_polynomiale <- lm(mpg ~ I(dis2) + disp, data = mtcars)  
summary(regression_polynomiale)
```

Call:

```
lm(formula = mpg ~ I(dis2) + disp, data = mtcars)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.9112	-1.5269	-0.3124	1.3489	5.3946

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.583e+01	2.209e+00	16.221	4.39e-16 ***
I(dis ²)	1.255e-04	3.891e-05	3.226	0.0031 **
disp	-1.053e-01	2.028e-02	-5.192	1.49e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.837 on 29 degrees of freedom

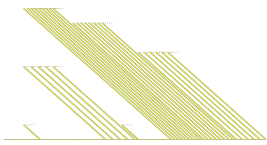
Multiple R-squared: 0.7927, Adjusted R-squared: 0.7784

F-statistic: 55.46 on 2 and 29 DF, p-value: 1.229e-10

vérification normalité des résidus

```
shapiro.test(regression_polynomiale$residuals)
```

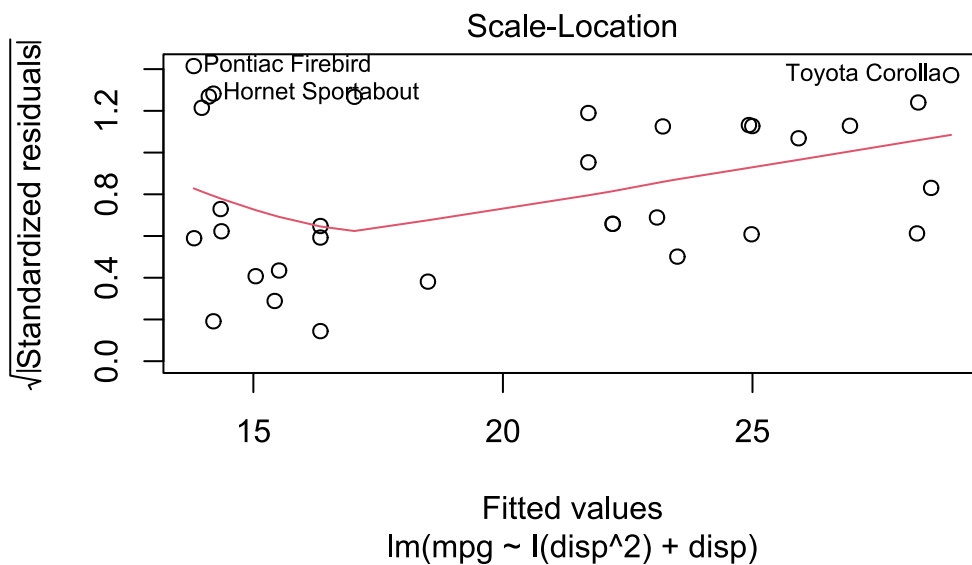
Shapiro-Wilk normality test



```
data: regression_polynomiale$residuals
W = 0.93679, p-value = 0.06073
```

vérification homogénéité

```
plot(regression_polynomiale, which = 3)
```



3.4 comparaison des deux modèles

```
AIC(regression_lineaire_cars, regression_polynomiale)
```

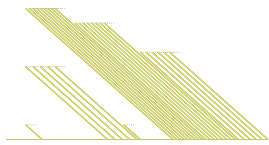
	df	AIC
regression_lineaire_cars	3	170.2094
regression_polynomiale	4	162.3957

```
anova(regression_lineaire_cars, regression_polynomiale)
```

Analysis of Variance Table

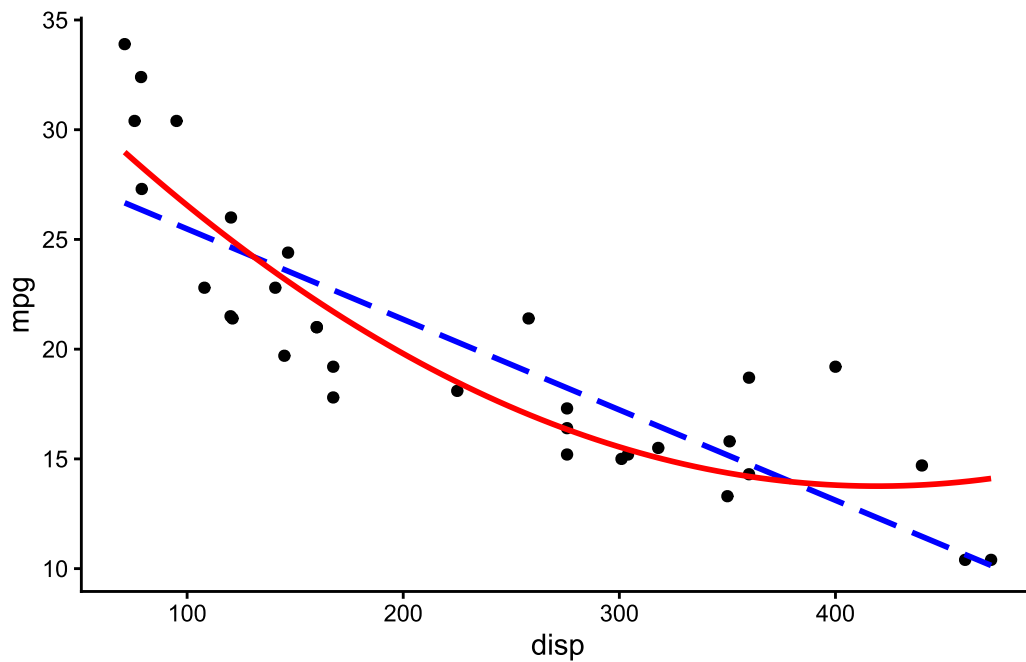
```
Model 1: mpg ~ disp
Model 2: mpg ~ I(disp^2) + disp
```

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
--------	-----	----	-----------	---	--------



```
1      30 317.16
2      29 233.39 1      83.766 10.408 0.003104 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ggplot(mtcars) +
  aes(y = mpg, x = disp) +
  geom_point() +
  geom_smooth(method = lm, formula = y ~ x, colour="blue", se = FALSE, linetype
= "longdash") +
  geom_smooth(method = lm, formula = y ~ x + I(x^2), color = "red", se = FALSE) +
  theme_classic()
```



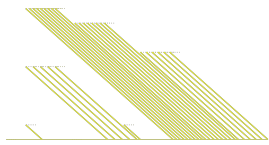
```
ggsave("img/04Rl16N.png", width = 14, height = 7, units = "cm")
```

4 Régression exponentielle

4.1 graphique

```
intervalle <- seq(-2.5, 3.5, 0.01)

tibble(
```

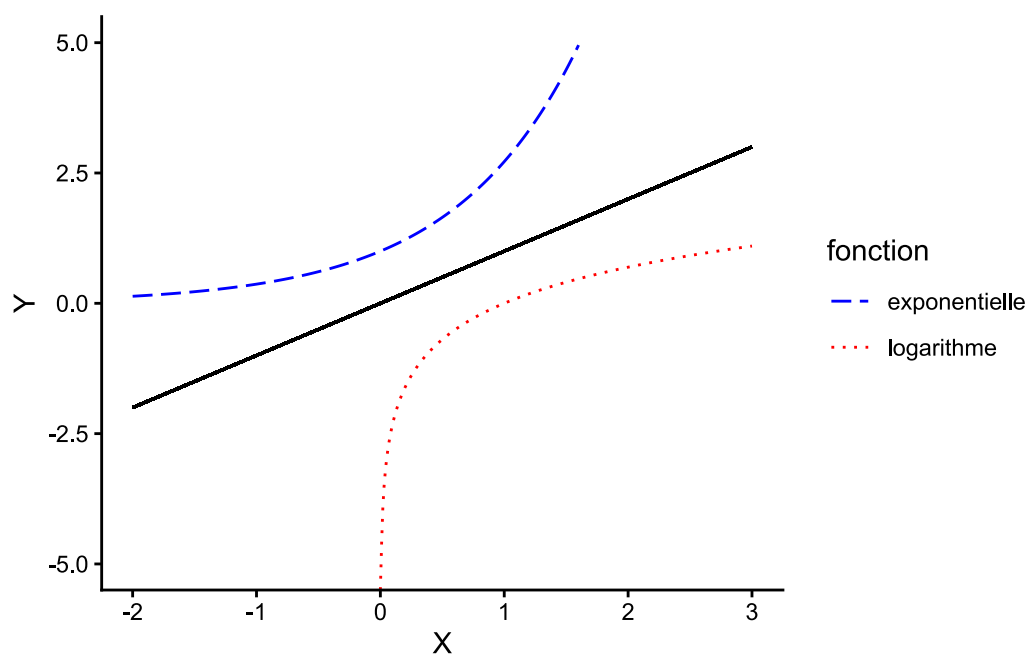
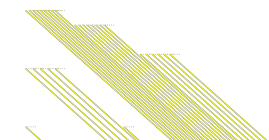


```
X = rep(intervalle, 2),
Y = c(exp(intervalle), log(intervalle)),
fonction = rep(c("exponentielle", "logarithme"), each = length(intervalle))
)|>
ggplot() +
  aes(x = X,
      y = Y,
      color = fonction,
      linetype = fonction
  ) +
  geom_line() +
  geom_segment(
    aes(x = -2, xend = 3, y = -2, yend = 3),
    color = "black",
    linetype = "solid"
  ) +
  xlim(-2, 3) +
  ylim(-5, 5) +
  scale_colour_manual(values = c("blue", "red")) +
  scale_linetype_manual(values = c("longdash", "dotted")) +
  theme_classic()
```

Warning in log(intervalle): Production de NaN

Warning in geom_segment(aes(x = -2, xend = 3, y = -2, yend = 3), color = "black", :
All aesthetics have length 1, but the data has 1202 rows.
i Please consider using `annotate()` or provide this layer with data containing
a single row.

Warning: Removed 540 rows containing missing values or values outside the scale
range
(`geom\_line()`).



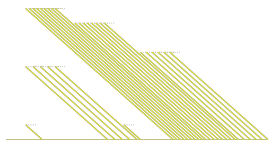
```
ggsave("img/04Rl17N.png", width = 14, height = 7, units = "cm")
```

```
Warning in geom_segment(aes(x = -2, xend = 3, y = -2, yend = 3), color = "black", :  
All aesthetics have length 1, but the data has 1202 rows.  
i Please consider using `annotate()` or provide this layer with data containing  
a single row.  
Removed 540 rows containing missing values or values outside the scale range  
(`geom_line()`).
```

4.2 régression exponentielle

```
regression_exponentielle <-  
  lm(  
    Ozone ~ exp(Temp),  
    data = airquality |>  
      filter(Temp < 86)  
  )  
  
regression_exponentielle |>  
  summary()
```

Call:



```
lm(formula = Ozone ~ exp(Temp), data = filter(airquality, Temp <
86))
```

Residuals:

Min	1Q	Median	3Q	Max
-40.087	-14.000	-6.643	6.362	139.493

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.764e+01	2.828e+00	9.772	1.18e-15 ***
exp(Temp)	5.770e-36	1.533e-36	3.765	0.000302 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 25.42 on 87 degrees of freedom

(30 observations effacées parce que manquantes)

Multiple R-squared: 0.1401, Adjusted R-squared: 0.1302

F-statistic: 14.17 on 1 and 87 DF, p-value: 0.0003023

vérification de la normalité

```
shapiro.test(regression_exponentielle$residuals)
```

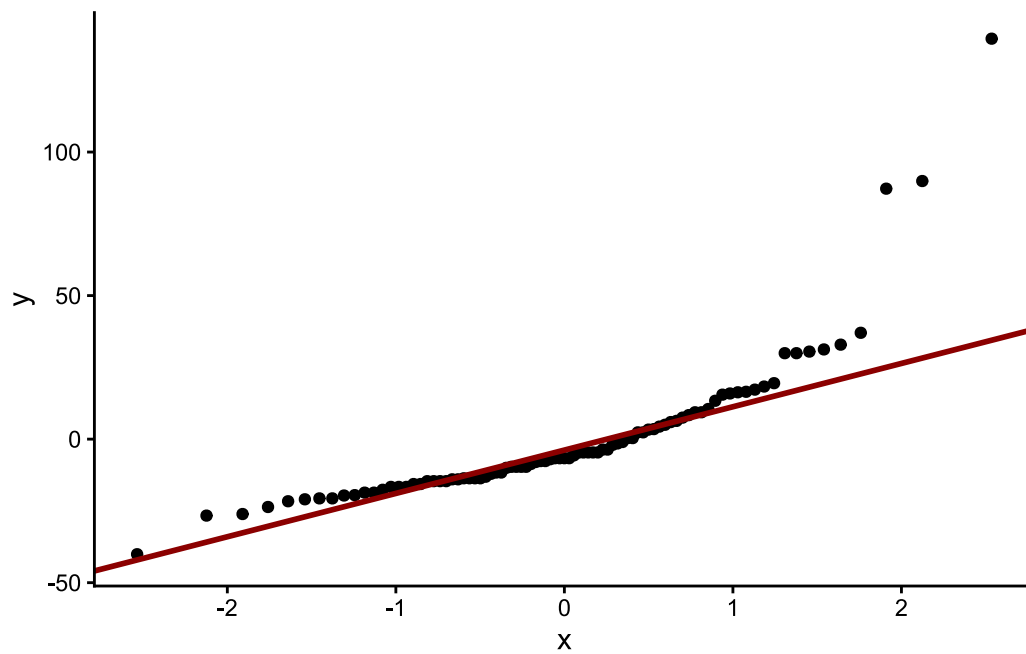
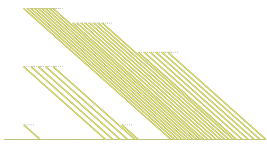
Shapiro-Wilk normality test

data: regression_exponentielle\$residuals

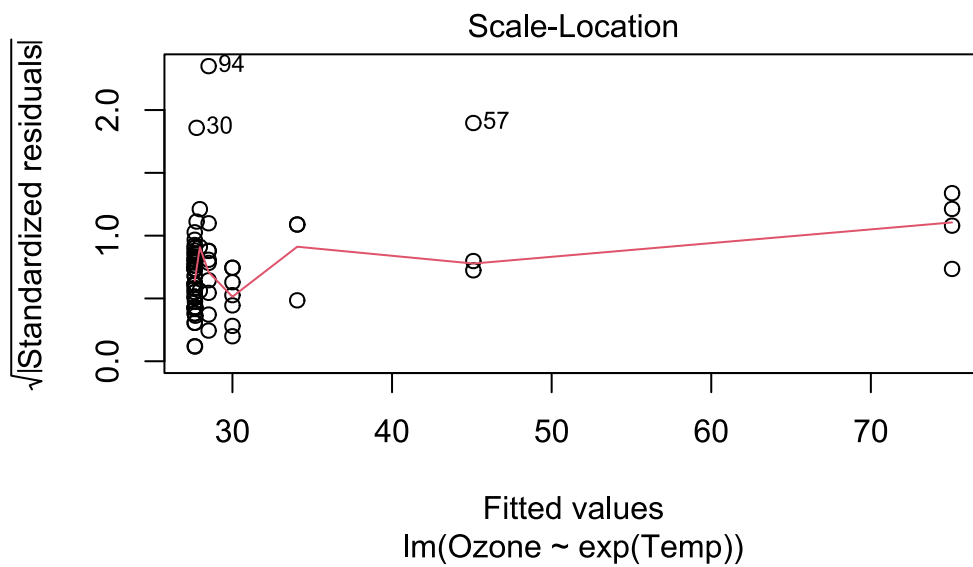
W = 0.73239, p-value = 1.932e-11

confirmation par un graphique

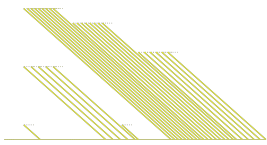
```
tibble(
  residus = regression_exponentielle$residuals
) |>
ggplot() +
  aes(sample = residus) +
  geom_qq() +
  geom_qq_line(color = "dark red", linewidth = 1) +
  theme_classic()
```



```
plot(regression_exponentielle, which = 3)
```



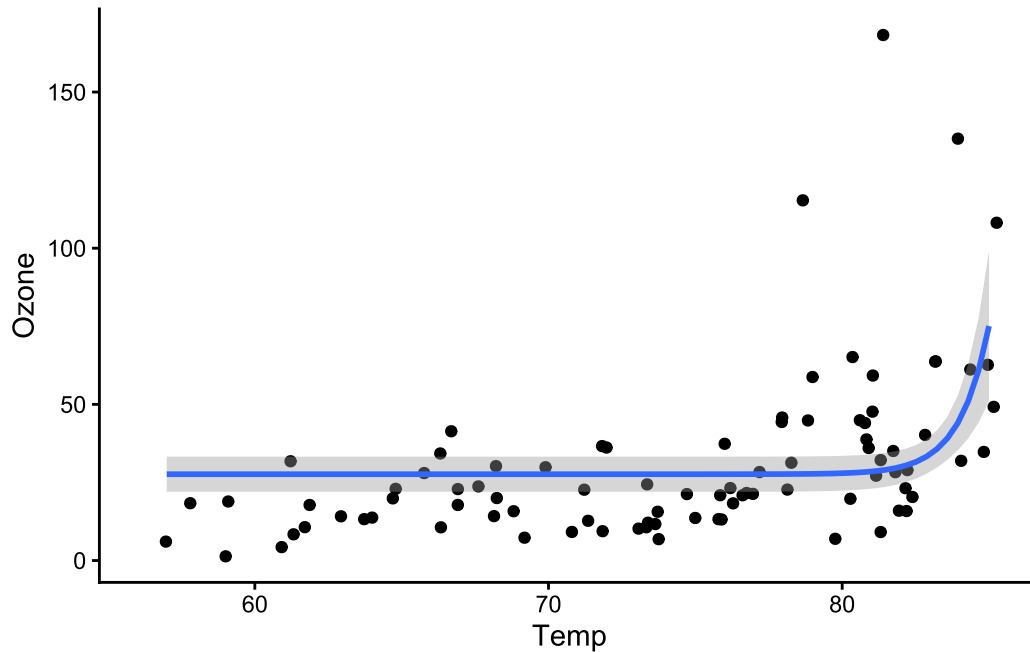
```
airquality |>
  filter(Temp < 86) |>
```



```
ggplot() +  
  aes(x = Temp, y = Ozone) +  
  geom_jitter() +  
  geom_smooth(method = lm, formula = y ~ exp(x)) +  
  theme_classic()
```

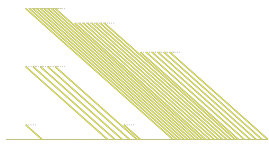
Warning: Removed 30 rows containing non-finite outside the scale range
(`stat\_smooth()`).

Warning: Removed 30 rows containing missing values or values outside the scale range
(`geom\_point()`).



5 Régression logistique binomiale

```
regression_logistique <- glm(  
  am ~ wt,  
  data = mtcars,  
  family = "binomial"  
)  
  
summary(regression_logistique)
```



```
Call:
glm(formula = am ~ wt, family = "binomial", data = mtcars)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   12.040      4.510    2.670  0.00759 **
wt            -4.024      1.436   -2.801  0.00509 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 43.230  on 31  degrees of freedom
Residual deviance: 19.176  on 30  degrees of freedom
AIC: 23.176

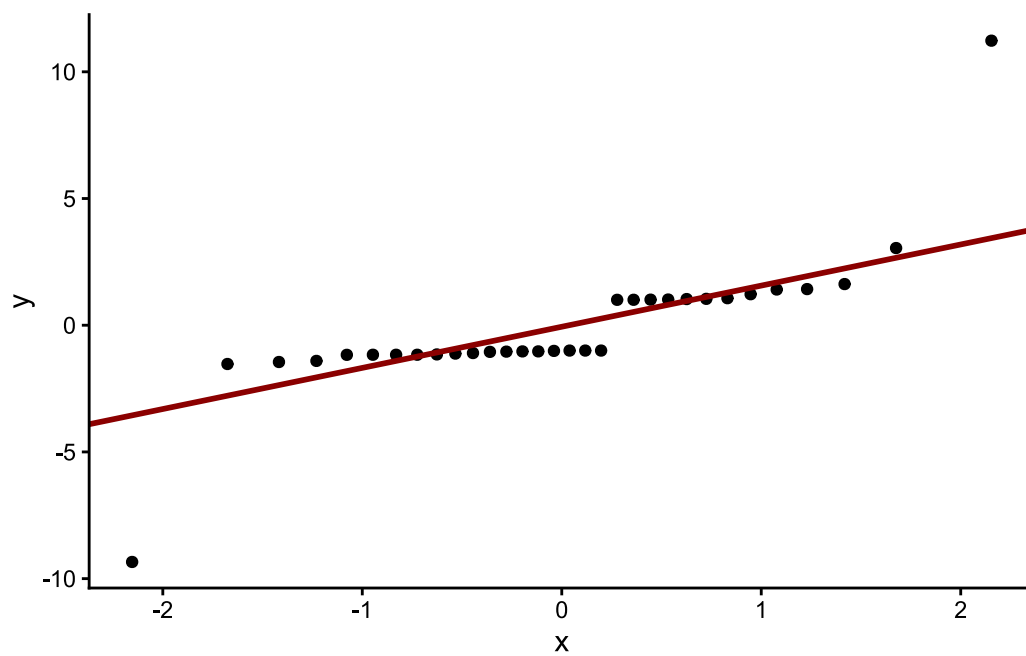
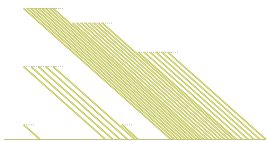
Number of Fisher Scoring iterations: 6
```

```
shapiro.test(regression_logistique$residuals)
```

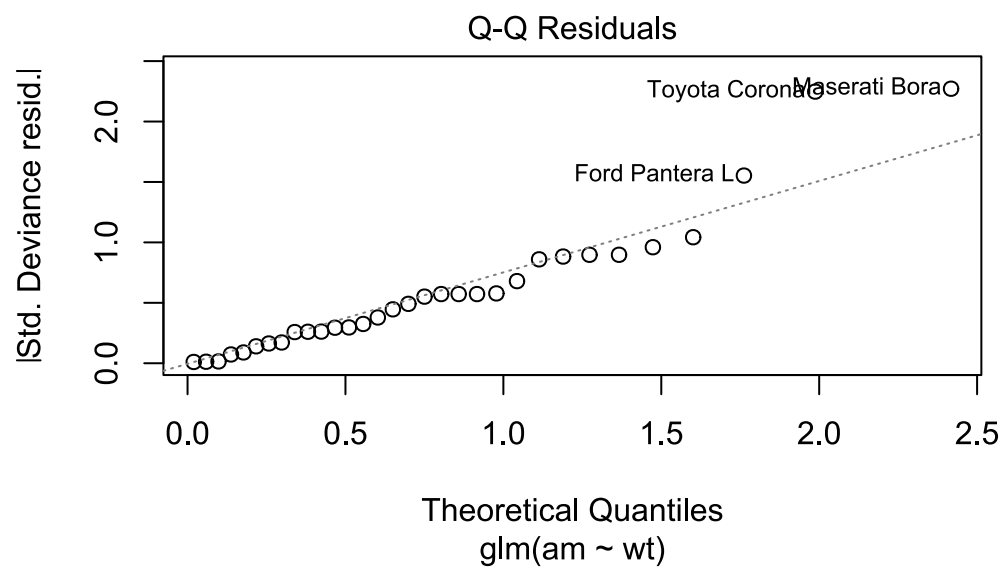
Shapiro-Wilk normality test

```
data:  regression_logistique$residuals
W = 0.69219, p-value = 6.658e-07
```

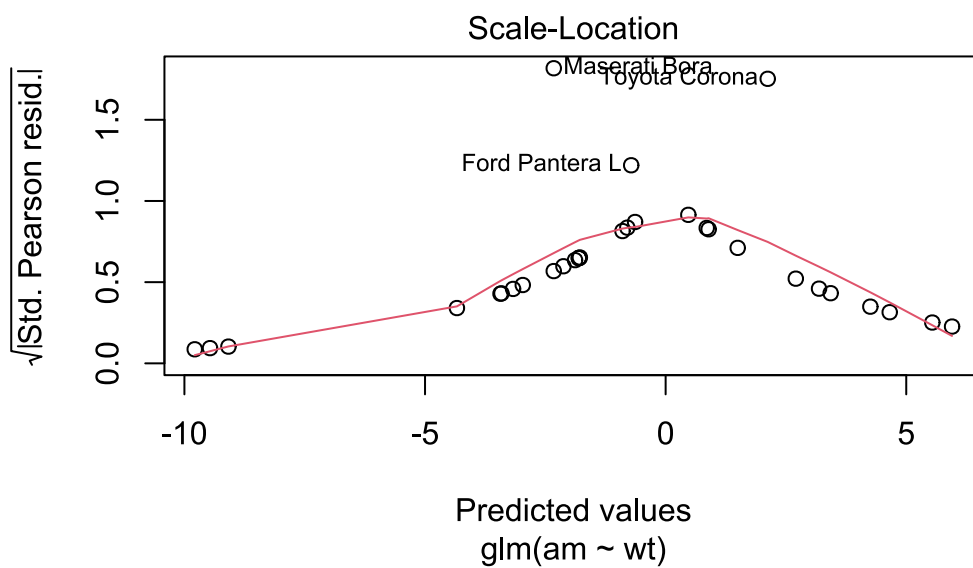
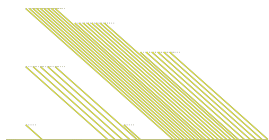
```
tibble(
  residus = regression_logistique$residuals
) |>
  ggplot() +
  aes(sample = residus) +
  geom_qq() +
  geom_qq_line(color = "dark red", size = 1) +
  theme_classic()
```



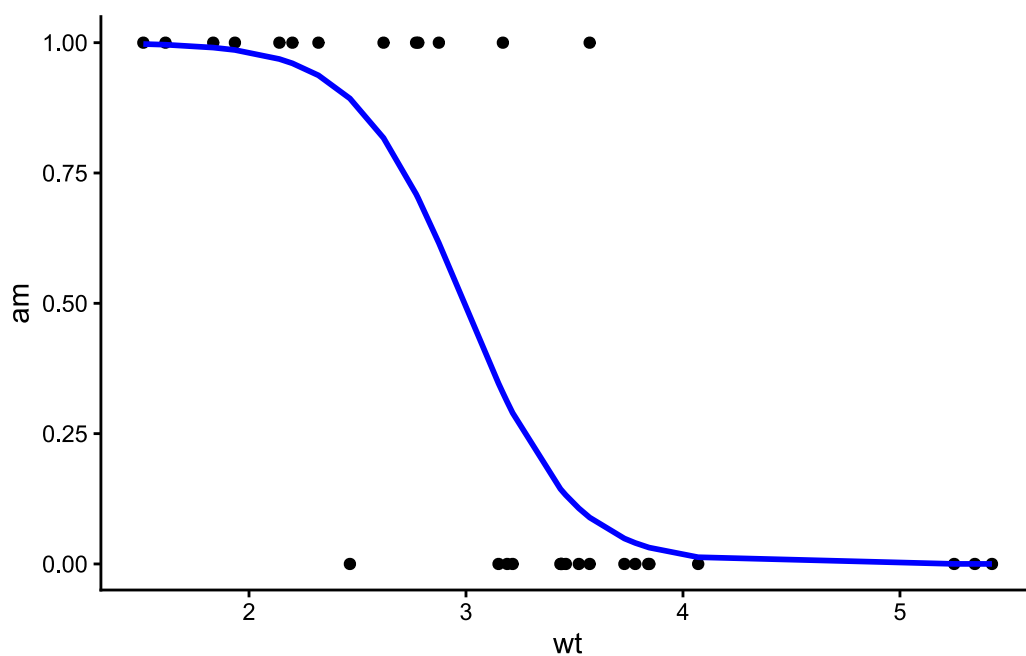
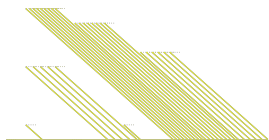
```
plot(regression_logistique, which = 2)
```



```
plot(regression_logistique, which = 3)
```

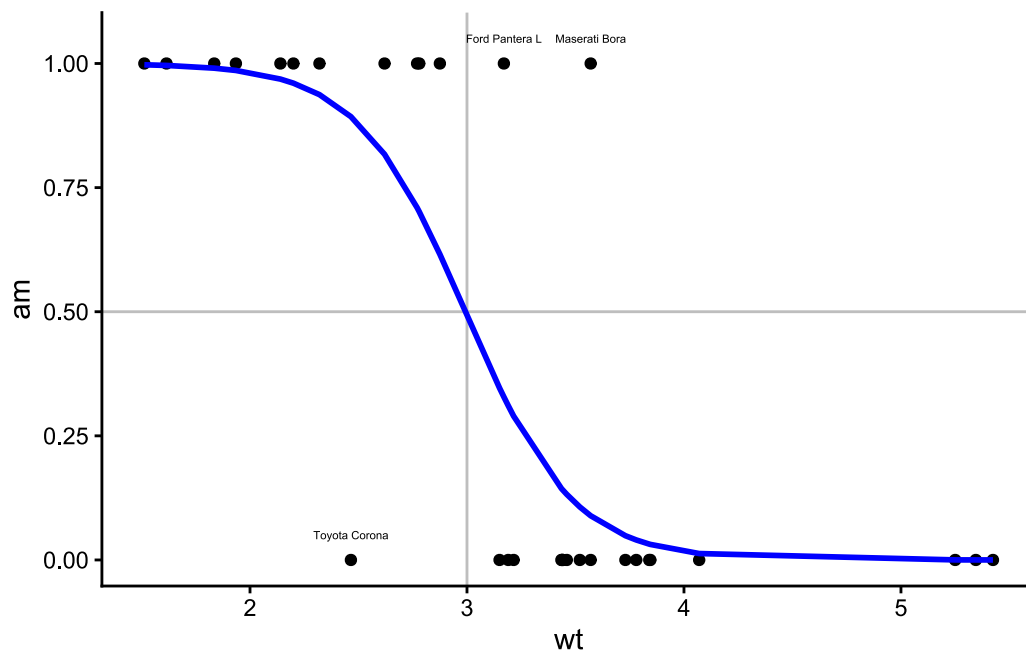
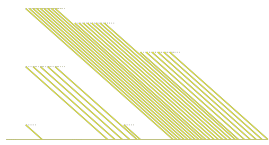


```
# version simplifiée
mtcars |>
  mutate(
    prediction_am = regression_logistique$fitted.values
  ) |>
  ggplot() +
  aes(x = wt, y = am) +
  geom_point() +
  geom_line(aes(y = prediction_am), color = "blue", linewidth = 1) +
  theme_classic()
```



```
# version annotée
mtcars |>
  rownames_to_column() |>
  mutate(
    prediction_am = regression_logistique$fitted.values,
    etiquette = case_when(
      rowname %in% c("Toyota Corona", "Maserati Bora", "Ford Pantera L") ~ rowname
    )
  ) |>
  ggplot() +
  aes(x = wt, y = am) +
  geom_point() +
  geom_hline(yintercept = 0.5, color = "grey") +
  geom_vline(xintercept = 3, color = "grey") +
  geom_line(aes(y = prediction_am), color = "blue", linewidth = 1) +
  geom_text(aes(label = etiquette), size = 1.5, nudge_y = 0.05) +
  theme_classic()
```

```
Warning: Removed 29 rows containing missing values or values outside the scale
range
(`geom_text()`).
```



6 Régression multiple

6.1 Régression multiple avec espèce

```
regression_multiple_espece <- lm(
  formula = Petal.Width ~ Petal.Length * Species,
  data = iris
)
```

```
regression_multiple_espece |>
  summary()
```

Call:

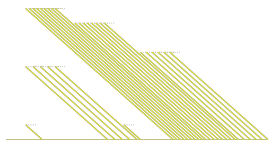
```
lm(formula = Petal.Width ~ Petal.Length * Species, data = iris)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.6337	-0.0744	-0.0134	0.0866	0.4503

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.04822	0.21472	-0.225	0.822627
Petal.Length	0.20125	0.14586	1.380	0.169813



```
Speciesversicolor      -0.03607    0.31538  -0.114  0.909109
Speciesvirginica        1.18425    0.33417   3.544  0.000532 ***
Petal.Length:Speciesversicolor  0.12981    0.15550   0.835  0.405230
Petal.Length:Speciesvirginica -0.04095    0.15291  -0.268  0.789244
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1773 on 144 degrees of freedom
Multiple R-squared:  0.9477,    Adjusted R-squared:  0.9459
F-statistic: 521.9 on 5 and 144 DF,  p-value: < 2.2e-16
```

```
car::Anova(regression_multiple_espece, type = "3")
```

Anova Table (Type III tests)

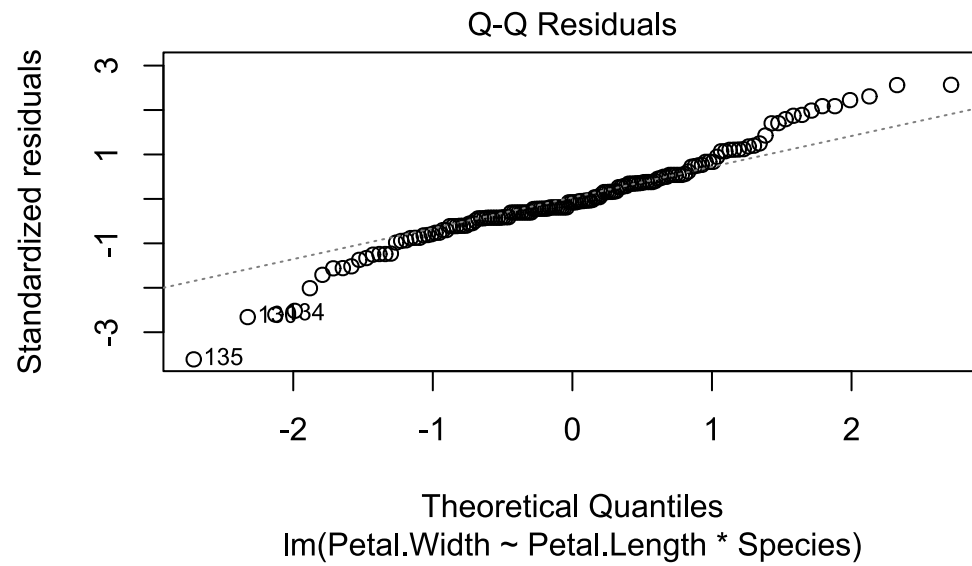
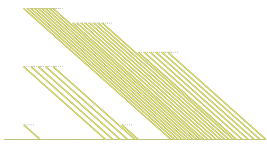
Response: Petal.Width

	Sum Sq	Df	F value	Pr(>F)
(Intercept)	0.0016	1	0.0504	0.8226271
Petal.Length	0.0599	1	1.9036	0.1698135
Species	0.5026	2	7.9931	0.0005107 ***
Petal.Length:Species	0.1842	2	2.9297	0.0566046 .
Residuals	4.5274	144		

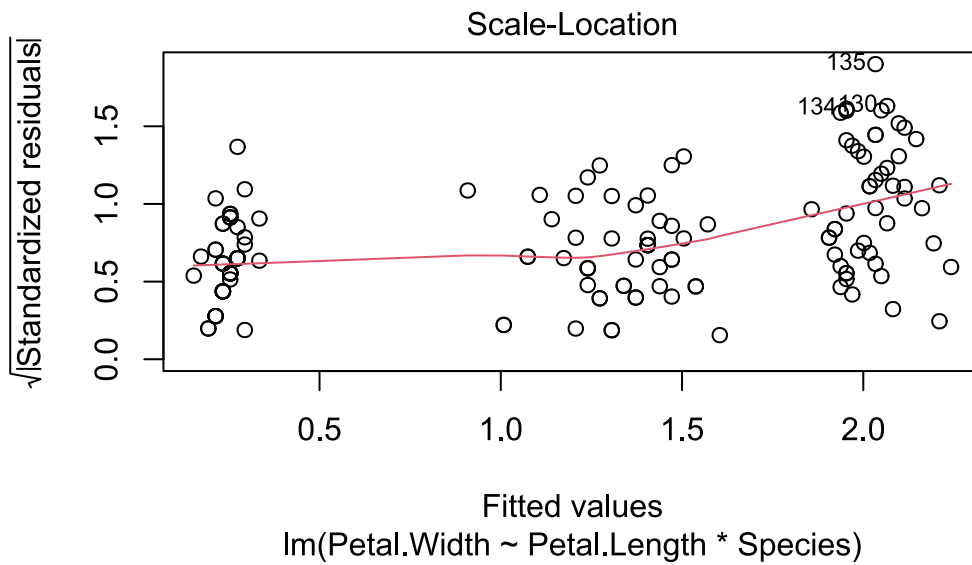
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

vérification du modèle avec espèce

```
plot(regression_multiple_espece, which = 2)
```

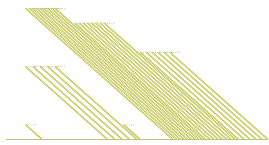


```
plot(regression_multiple_espece, which = 3)
```



test post-hoc

```
library(multcomp)
```



Le chargement a nécessité le package : mvtnorm

Le chargement a nécessité le package : survival

Le chargement a nécessité le package : TH.data

Le chargement a nécessité le package : MASS

Attachement du package : 'MASS'

L'objet suivant est masqué depuis 'package:dplyr':

select

Attachement du package : 'TH.data'

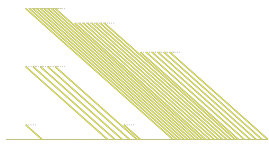
L'objet suivant est masqué depuis 'package:MASS':

geyser

```
summary(  
  glht(  
    regression_multiple_espece,  
    linfct = mcp(  
      Species = "Tukey"  
    )  
  )  
)
```

Warning in mcp2matrix(model, linfct = linfct): covariate interactions found -- default contrast might be inappropriate

Simultaneous Tests for General Linear Hypotheses



Multiple Comparisons of Means: Tukey Contrasts

Fit: `lm(formula = Petal.Width ~ Petal.Length * Species, data = iris)`

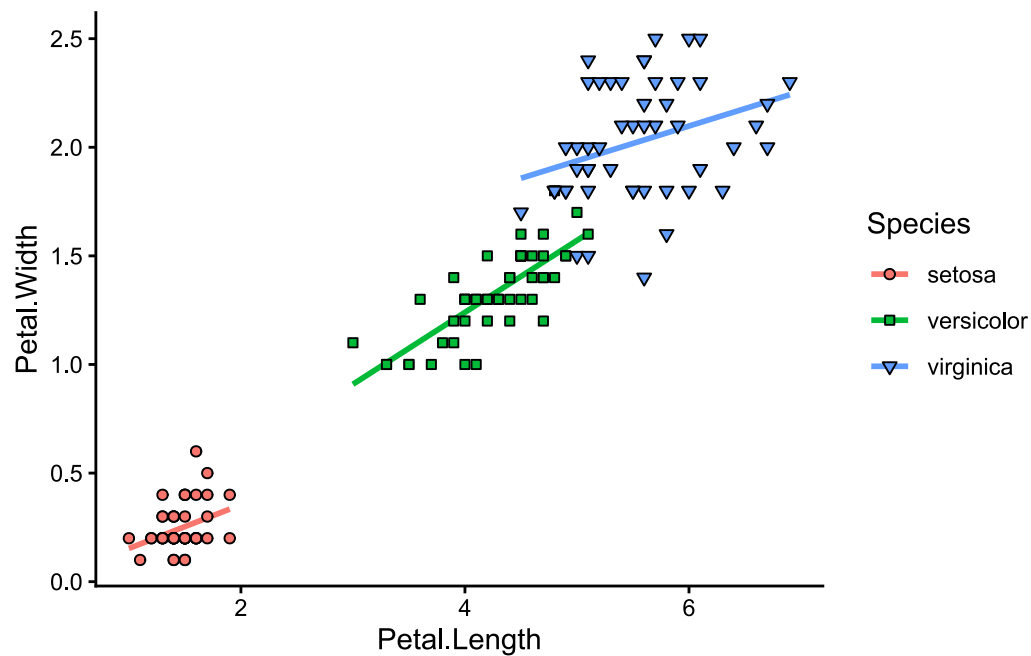
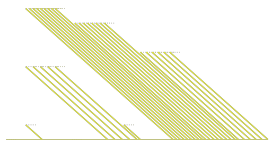
Linear Hypotheses:

	Estimate	Std. Error	t value	Pr(> t)
versicolor - setosa == 0	-0.03607	0.31538	-0.114	0.9928
virginica - setosa == 0	1.18425	0.33417	3.544	0.0015 **
virginica - versicolor == 0	1.22032	0.34486	3.539	0.0015 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Adjusted p values reported -- single-step method)

```
ggplot(iris) +  
  aes(  
    x = Petal.Length,  
    y = Petal.Width,  
    color = Species,  
    fill = Species,  
    shape = Species  
  ) +  
  geom_smooth(method = lm, se = FALSE) +  
  geom_point(color = "black") +  
  scale_shape_manual(values = c(21, 22, 25)) +  
  theme_classic()
```

``geom_smooth()`` using formula = 'y ~ x'



```
ggsave("img/04RL32N.png", width = 14, height = 7, units = "cm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

6.2 modèle initial, régression multiple

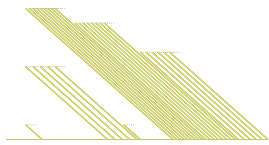
```
regression_multiple <- lm(  
  formula = Petal.Width ~ Petal.Length + I(Sepal.Length^2) + Sepal.Length *  
  exp(Sepal.Width),  
  data = iris  
)
```

```
car::Anova(regression_multiple, type = "3")
```

Anova Table (Type III tests)

Response: Petal.Width

	Sum Sq	Df	F value	Pr(>F)
(Intercept)	0.0654	1	1.7179	0.19205
Petal.Length	12.4346	1	326.5651	< 2e-16 ***
I(Sepal.Length^2)	0.1266	1	3.3243	0.07034 .
Sepal.Length	0.0413	1	1.0840	0.29954
exp(Sepal.Width)	0.0071	1	0.1874	0.66575



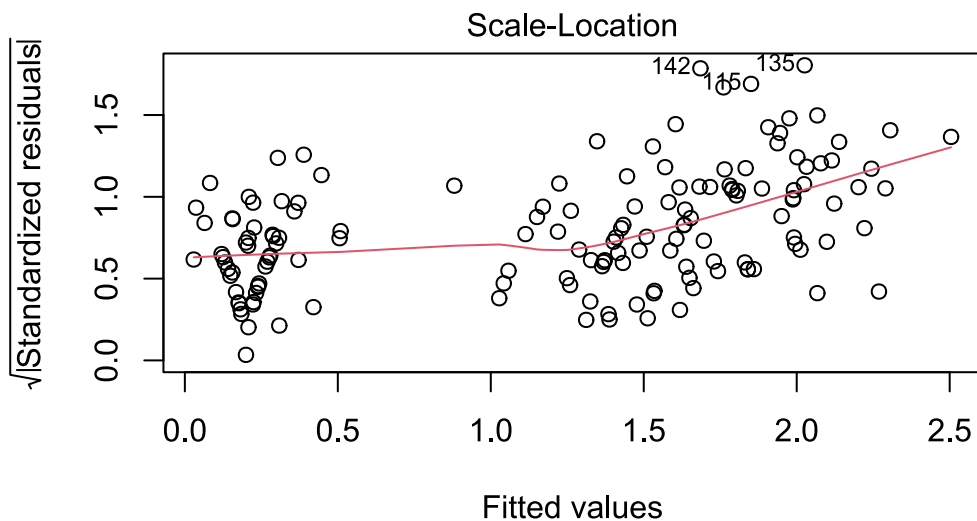
```
Sepal.Length:exp(Sepal.Width) 0.0342 1 0.8974 0.34506
Residuals                    5.4831 144
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
shapiro.test(regression_multiple$residuals)
```

Shapiro-Wilk normality test

```
data:  regression_multiple$residuals
W = 0.98367, p-value = 0.07271
```

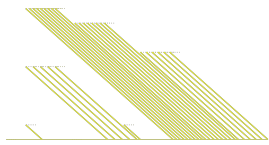
```
plot(regression_multiple, which = 3)
```



```
lm(Petal.Width ~ Petal.Length + I(Sepal.Length^2) + Sepal.Length * exp(Sepal.Width))
```

deuxième modèle

```
m2 <- lm(
  formula = Petal.Width ~ Petal.Length + I(Sepal.Length^2) + Sepal.Length +
  exp(Sepal.Width),
  data = iris
)
```



```
car::Anova(m2, type = "3")
```

Anova Table (Type III tests)

Response: Petal.Width

	Sum Sq	Df	F value	Pr(>F)
(Intercept)	0.0672	1	1.7656	0.1860137
Petal.Length	14.1860	1	372.8273	< 2.2e-16 ***
I(Sepal.Length^2)	0.0981	1	2.5776	0.1105631
Sepal.Length	0.0305	1	0.8019	0.3719993
exp(Sepal.Width)	0.4758	1	12.5049	0.0005454 ***
Residuals	5.5172	145		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
anova(regression_multiple, m2)
```

Analysis of Variance Table

Model 1: Petal.Width ~ Petal.Length + I(Sepal.Length^2) + Sepal.Length *
exp(Sepal.Width)

Model 2: Petal.Width ~ Petal.Length + I(Sepal.Length^2) + Sepal.Length +
exp(Sepal.Width)

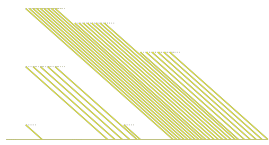
	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	144	5.4831				
2	145	5.5172	-1	-0.03417	0.8974	0.3451

```
AIC(regression_multiple, m2)
```

	df	AIC
regression_multiple	7	-56.66406
m2	6	-57.73216

troisième modèle

```
m3 <- lm(  
  formula = Petal.Width ~ Petal.Length + I(Sepal.Length^2) + exp(Sepal.Width),  
  data = iris  
)
```



```
car::Anova(m3, type = "3")
```

Anova Table (Type III tests)

Response: Petal.Width

	Sum Sq	Df	F value	Pr(>F)
(Intercept)	0.9337	1	24.571	1.961e-06 ***
Petal.Length	17.9475	1	472.324	< 2.2e-16 ***
I(Sepal.Length^2)	0.6303	1	16.587	7.592e-05 ***
exp(Sepal.Width)	0.5502	1	14.479	0.0002079 ***
Residuals	5.5478	146		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
anova(m3, m2)
```

Analysis of Variance Table

Model 1: Petal.Width ~ Petal.Length + I(Sepal.Length^2) + exp(Sepal.Width)

Model 2: Petal.Width ~ Petal.Length + I(Sepal.Length^2) + Sepal.Length +
exp(Sepal.Width)

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	146	5.5478				
2	145	5.5172	1	0.030514	0.8019	0.372

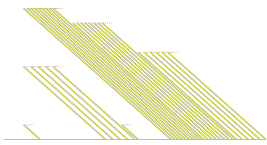
```
AIC(m3, m2)
```

	df	AIC
m3	5	-58.90486
m2	6	-57.73216

validation du modèle final

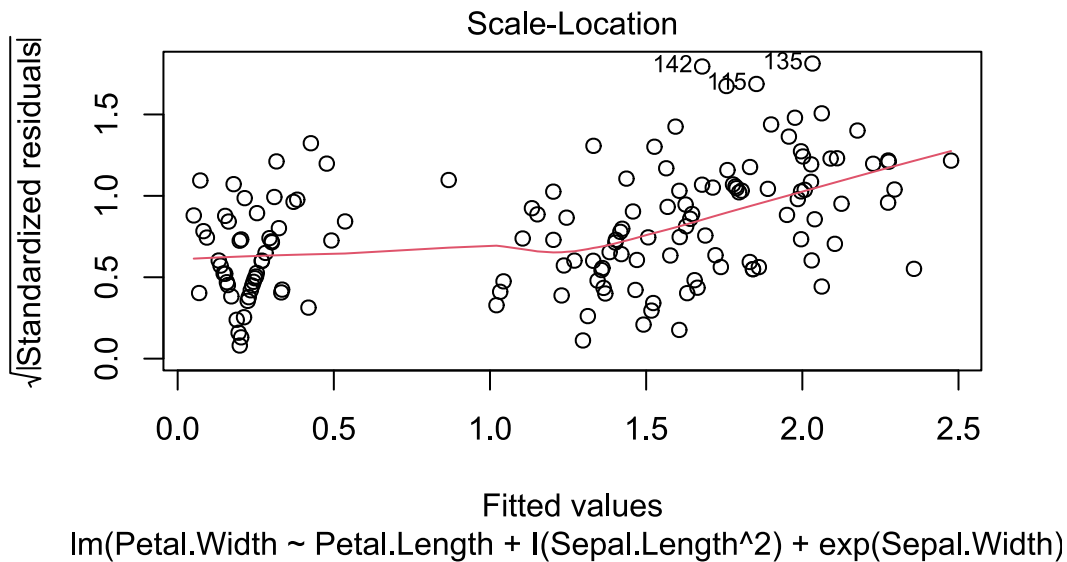
```
shapiro.test(m3$residuals)
```

Shapiro-Wilk normality test



```
data: m3$residuals  
W = 0.98388, p-value = 0.07691
```

```
plot(m3, which = 3)
```

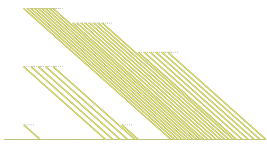


7 Regression non paramétrique univariée

7.1 dualité biais-variance

```
ggplot(airquality) +  
  aes(x = Temp, y = Ozone) +  
  geom_point(color = "grey") +  
  geom_smooth(method = "loess", linetype = "twodash", span = 0.2, color = "red",  
    se = FALSE) +  
  geom_smooth(method = "loess", linetype = "dotdash", span = 2, color = "green",  
    se = FALSE) +  
  geom_smooth(method = "loess", span = 0.75, color = "blue", se = FALSE) +  
  theme_classic()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

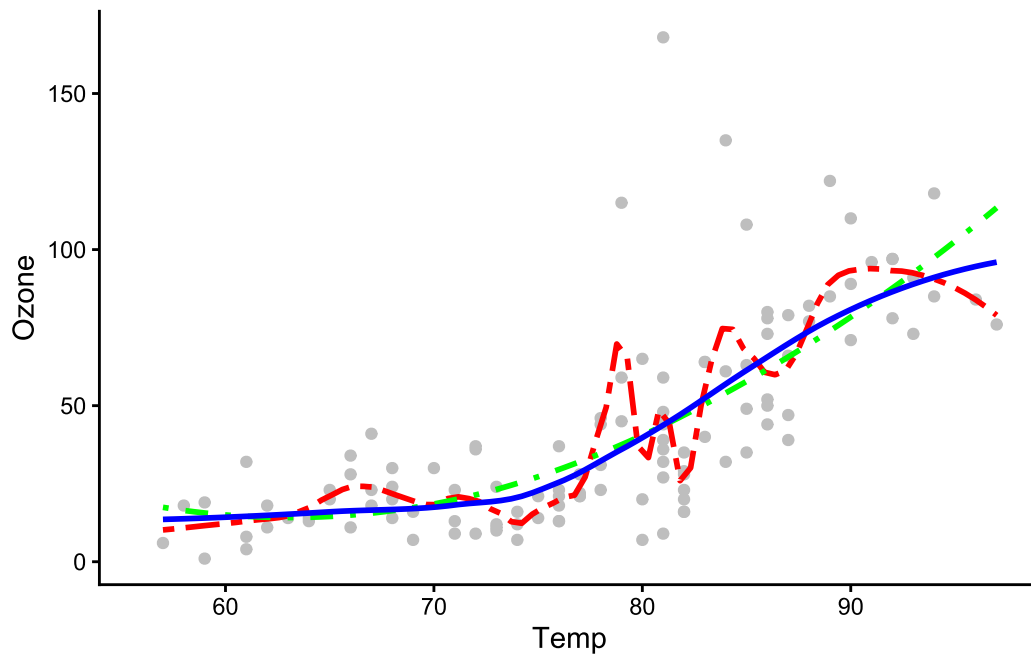
```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
`geom_smooth()` using formula = 'y ~ x'
```

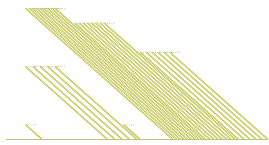
```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 37 rows containing missing values or values outside the scale
range
(`geom_point()`).
```



```
ggsave("img/04RL39N.png", width = 14, height = 7, units = "cm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_smooth()`).
Removed 37 rows containing missing values or values outside the scale range
(`geom_point()`).
```

7.2 Ozone ~ Temp

```
library(gam)
```

Le chargement a nécessité le package : splines

Le chargement a nécessité le package : foreach

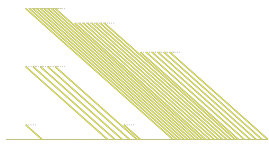
Attachement du package : 'foreach'

Les objets suivants sont masqués depuis 'package:purrr':

accumulate, when

Loaded gam 1.22-7

```
gam_uni <- gam(
  Ozone ~ lo(Temp),
  data = airquality
)
```



```
summary(gam_uni)
```

```
Call: gam(formula = Ozone ~ lo(Temp), data = airquality)
Deviance Residuals:
    Min       1Q   Median       3Q      Max
-34.999 -11.204  -3.733   8.843 124.001

(Dispersion Parameter for gaussian family taken to be 487.0196)

Null Deviance: 125143.1 on 115 degrees of freedom
Residual Deviance: 54285.01 on 111.4637 degrees of freedom
AIC: 1053.482
37 observations effacées parce que manquantes

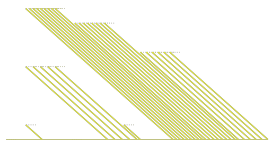
Number of Local Scoring Iterations: NA

Anova for Parametric Effects
              Df Sum Sq Mean Sq F value    Pr(>F)
lo(Temp)      1.00  61033    61033  125.32 < 2.2e-16 ***
Residuals    111.46  54285         487
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Anova for Nonparametric Effects
              Npar Df Npar F      Pr(F)
(Intercept)
lo(Temp)      2.5   7.954 0.0001904 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
regression_exponentielle <- lm(Ozone ~ exp(Temp), data = airquality)
regression_polynomiale <- lm(Ozone ~ Temp + I(Temp^2), data = airquality)
AIC(gam_uni, regression_exponentielle, regression_polynomiale)
```

	df	AIC
gam_uni	3	1053.482
regression_exponentielle	3	1142.483
regression_polynomiale	4	1056.148



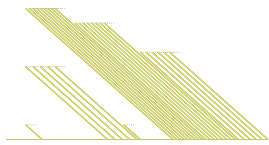
```
airquality |>
  rownames_to_column() |>
  full_join(
    predict(gam_uni) |>
      as.data.frame() |>
        rownames_to_column() |>
          rename(ozone_predit = "predict(gam_uni)")
  ) |>
  ggplot() +
  aes(x = Temp, y = Ozone) +
  geom_jitter() +
  geom_smooth(
    method = lm,
    formula = y ~ exp(x),
    data = airquality |> filter(Temp < 86),
    linetype = "twodash",
    se = FALSE
  ) +
  geom_smooth(
    method = lm,
    formula = y ~ x + I(x^2),
    color = "green",
    se = FALSE
  ) +
  geom_smooth(
    aes(y = ozone_predit),
    linetype = "longdash",
    color = "red"
  ) +
  theme_classic()
```

Joining with `by = join\_by(rowname)`

Warning: Removed 30 rows containing non-finite outside the scale range
(`stat\_smooth()`).

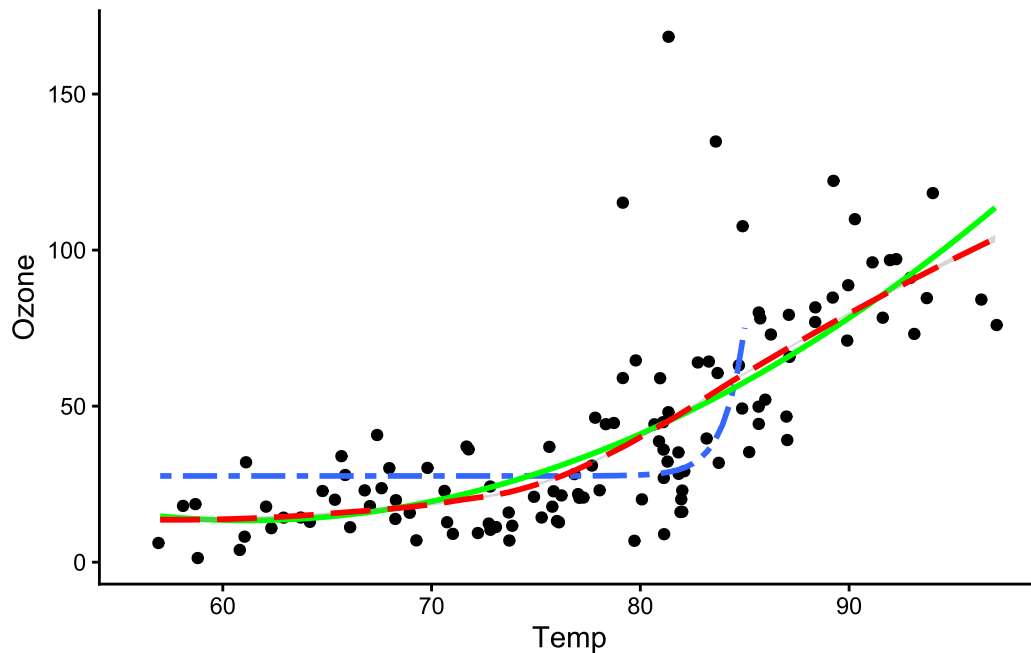
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat\_smooth()`).

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'



```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 37 rows containing missing values or values outside the scale
range
(`geom_point()`).
```



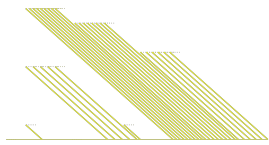
```
ggsave("img/04Rl42N.png", width = 14, height = 7, units = "cm")
```

```
Warning: Removed 30 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_smooth()`).
```



```
Warning: Removed 37 rows containing missing values or values outside the scale
range
(`geom_point()`).
```

7.3 regression_non_parametrique_multivariee

```
library(gam)

gam_initial <- gam(
  Ozone ~ lo(Temp) + lo(Wind) + lo(Solar.R) + lo(Day),
  data = airquality
)

summary(gam_initial)
```

```
Call: gam(formula = Ozone ~ lo(Temp) + lo(Wind) + lo(Solar.R) + lo(Day),
  data = airquality)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-43.074	-9.201	-2.155	7.836	65.410

(Dispersion Parameter for gaussian family taken to be 298.2742)

Null Deviance: 121801.9 on 110 degrees of freedom
Residual Deviance: 28469.42 on 95.4471 degrees of freedom
AIC: 963.8333
42 observations effacées parce que manquantes

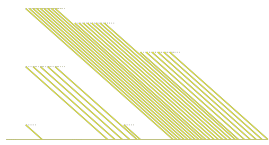
Number of Local Scoring Iterations: NA

Anova for Parametric Effects

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
lo(Temp)	1.000	47237	47237	158.3680	< 2.2e-16 ***
lo(Wind)	1.000	11164	11164	37.4270	2.082e-08 ***
lo(Solar.R)	1.000	3724	3724	12.4863	0.0006339 ***
lo(Day)	1.000	1102	1102	3.6952	0.0575512 .
Residuals	95.447	28469	298		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Anova for Nonparametric Effects



```
      Npar Df  Npar F      Pr(F)
(Intercept)
lo(Temp)      2.5  7.0290  0.000564 ***
lo(Wind)      2.8 10.9651  4.782e-06 ***
lo(Solar.R)   2.6  1.5648  0.208397
lo(Day)       2.6  1.1102  0.344659
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

gam_2

```
gam_2 <- gam(
  Ozone ~ lo(Temp) + lo(Wind) + lo(Solar.R),
  data = airquality
)

AIC(gam_initial, gam_2)
```

```
      df      AIC
gam_initial 6 963.8333
gam_2       5 963.6606
```

```
summary(gam_2)
```

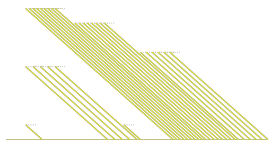
```
Call: gam(formula = Ozone ~ lo(Temp) + lo(Wind) + lo(Solar.R), data = airquality)
Deviance Residuals:
    Min       1Q   Median       3Q      Max
-46.417  -9.773  -3.465   8.943  71.975
```

(Dispersion Parameter for gaussian family taken to be 306.3018)

```
Null Deviance: 121801.9 on 110 degrees of freedom
Residual Deviance: 30348.83 on 99.0815 degrees of freedom
AIC: 963.6606
42 observations effacées parce que manquantes
```

Number of Local Scoring Iterations: NA

```
Anova for Parametric Effects
      Df Sum Sq Mean Sq F value    Pr(>F)
```

```
lo(Temp)      1.000  48471  48471 158.246 < 2.2e-16 ***
lo(Wind)      1.000  11113  11113  36.282 2.914e-08 ***
lo(Solar.R)   1.000   3472   3472  11.335 0.001084 **
Residuals    99.081  30349    306

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Anova for Nonparametric Effects
      Npar Df Npar F      Pr(F)
(Intercept)
lo(Temp)      2.5 6.9452 0.0006017 ***
lo(Wind)      2.8 9.9053 1.386e-05 ***
lo(Solar.R)   2.6 2.3287 0.0885503 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

gam_3

```
gam_3 <- gam(
  Ozone ~ lo(Temp) + lo(Wind),
  data = airquality
)
```

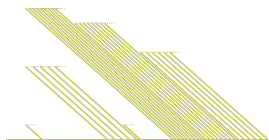
```
AIC(gam_2, gam_3)
```

Warning in AIC.default(gam_2, gam_3): tous les modèles n'ont pas été ajustés sur le même nombre d'observations

```
      df      AIC
gam_2  5  963.6606
gam_3  4 1019.0723
```

```
gam_3_bis <- gam(
  Ozone ~ lo(Temp) + lo(Wind),
  data = airquality |> filter(!is.na(Solar.R))
)
```

```
AIC(gam_2, gam_3_bis)
```



	df	AIC
gam_2	5	963.6606
gam_3_bis	4	972.6289